IPTA 2026 · INTERNATIONAL PULSAR TIMING ARRAY MEETING

Timing the Double Neutron Star PSR J1756−2251

Two decades of timing baseline, some questions answered, more questions asked.

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1 A laboratory for strong-field gravity

PSR J1756−2251 is a recycled pulsar in a relativistic double neutron star system with a 7.67-h, mildly eccentric orbit. We extend its timing baseline by six years through dense MeerKAT L/S-band and Parkes UWL observations, bringing the total span to over two decades. The extended data set yields improved measurements of five post-Keplerian parameters, tightening constraints on the component masses and the system's relativistic dynamics.

28.46 ms SPIN PERIOD	**7.67 h** ORBITAL PERIOD	**0.1805** ECCENTRICITY
1.34 M☉ PULSAR MASS	**1.23 M☉** COMPANION MASS	**2004–26** TIMING BASELINE

2 Two decades, 7 telescopes, 12 Backends

Our times of arrival span 2004–2026. The early data set comprises legacy campaigns with the Parkes, Westerbork, Jodrell Bank, Green Bank, and Nançay telescopes; these are complemented by dense L- and S-band monitoring with MeerKAT since 2019, Parkes UWL observations from 2021, and new Effelsberg epochs beginning in 2025. Profiles matched at 2.6 GHz with a bandwidth of 200MHz from Parkes, MeerKAT, and Effelsberg exhibit the same sharp main-pulse morphology, and frequency-resolved templates were constructed for each new backend for ToA generation.

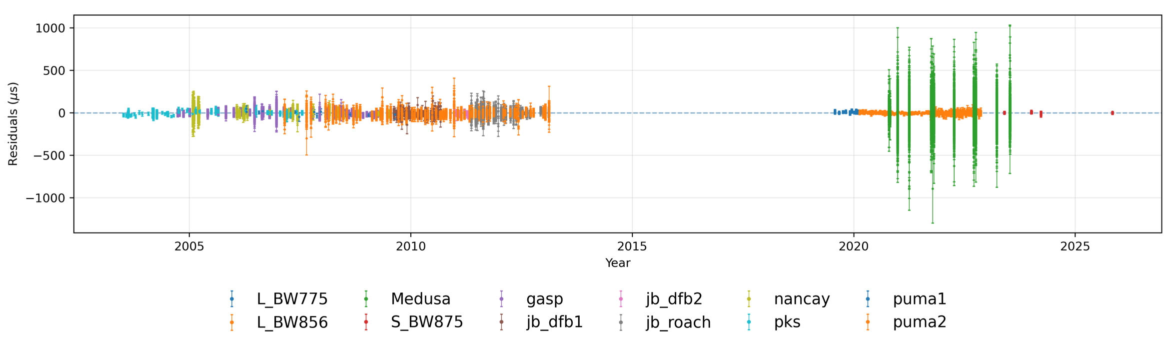


Fig. 1 *Timing residuals from 12 backends under the DDH timing model with the favoured white-noise, red-noise, and DM noise model.*

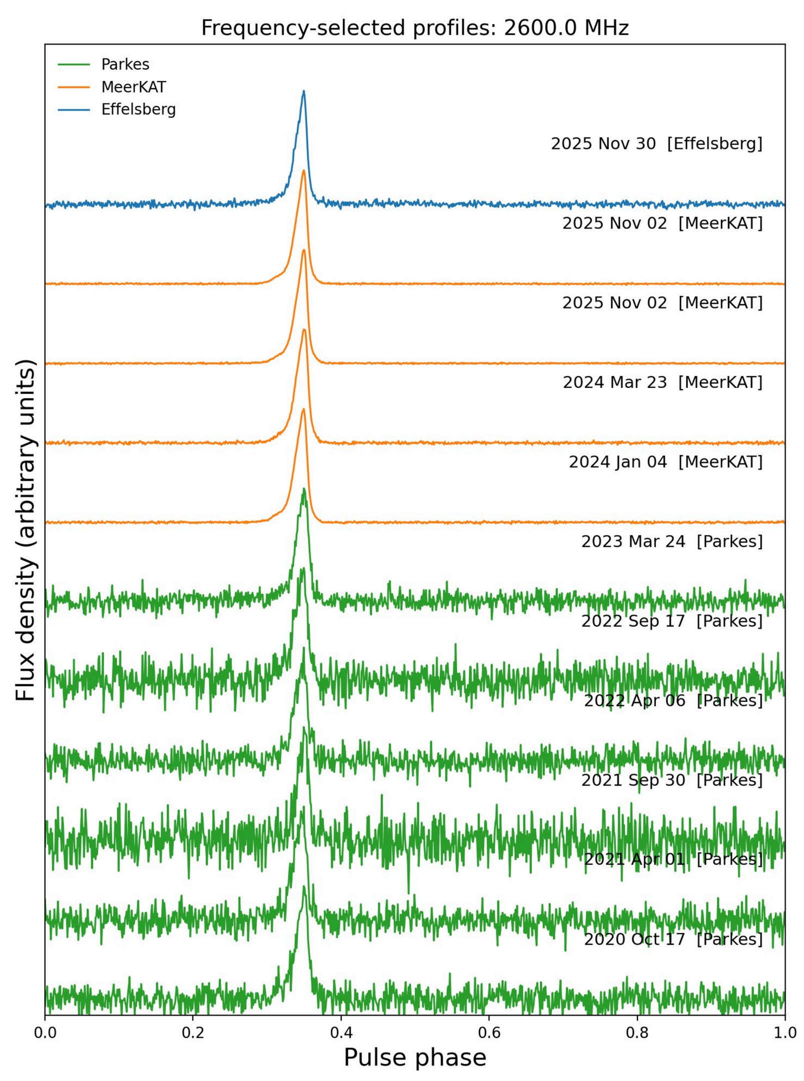


Fig. 2 *Profiles matched at 2.6 GHz with a bandwidth of 200MHz from Parkes (green), MeerKAT (orange), and Effelsberg (blue) spanning 2020–2025, showing a consistent, sharp main pulse across all three telescopes.*

3 Better constraints on everything!

Parameter	Value
Right ascension, RAJ (hms)	17:56:46.632944(47)
Declination, DECJ (dms)	−22:51:59.419(57)
Proper motion in RA, PMRA (mas/yr)	−2.015(79)
Proper motion in Dec, PMDEC (mas/yr)	9.9(7.0)
Parallax, PX (mas)	−2.02(1.66)
Spin frequency derivative, F1 (s^{-2})	$-1.255741(95)\times10^{-15}$
Dispersion measure, DM (cm^{-3} pc)	121.158(39)
Orbital period, PB (d)	0.31963390155609(34)
Projected semi-major axis, A1 (lt-s)	2.7564799(27)
Eccentricity, ECC	0.18056971(11)
Periastron advance, OMDOT (deg/yr)	2.5823837(50)
Einstein delay, GAMMA (s)	$1.1317(27)\times10^{-3}$
Orbital period derivative, PBDOT (s/s)	$-2.02693(25)\times10^{-13}$
Orthometric amplitude, H3 (s)	$1.611(44)\times10^{-6}$
Orthometric ratio, STIG	0.687(21)
Excess orbital decay, XPBDOT (s/s)	$1.3955(25)\times10^{-14}$
Companion mass, M2 (Msun)	1.2300(20)
Pulsar mass, Mp = MTOT − M2 (Msun)	1.3400(20)

Our extended 22-year dataset yields improved measurements of five PK parameters. The periastron advance $\dot{\omega}$ and Einstein delay γ are now measured with significantly reduced uncertainties, while the orbital period derivative $\dot{P}_b$ is detected at a fractional precision of $0.012\,\%$. The addition of the orthometric Shapiro-delay amplitude $h_3$and ratio ς, not significantly measured in the earlier study, now provides two independent Shapiro-delay constraints, tightening the component mass estimates.

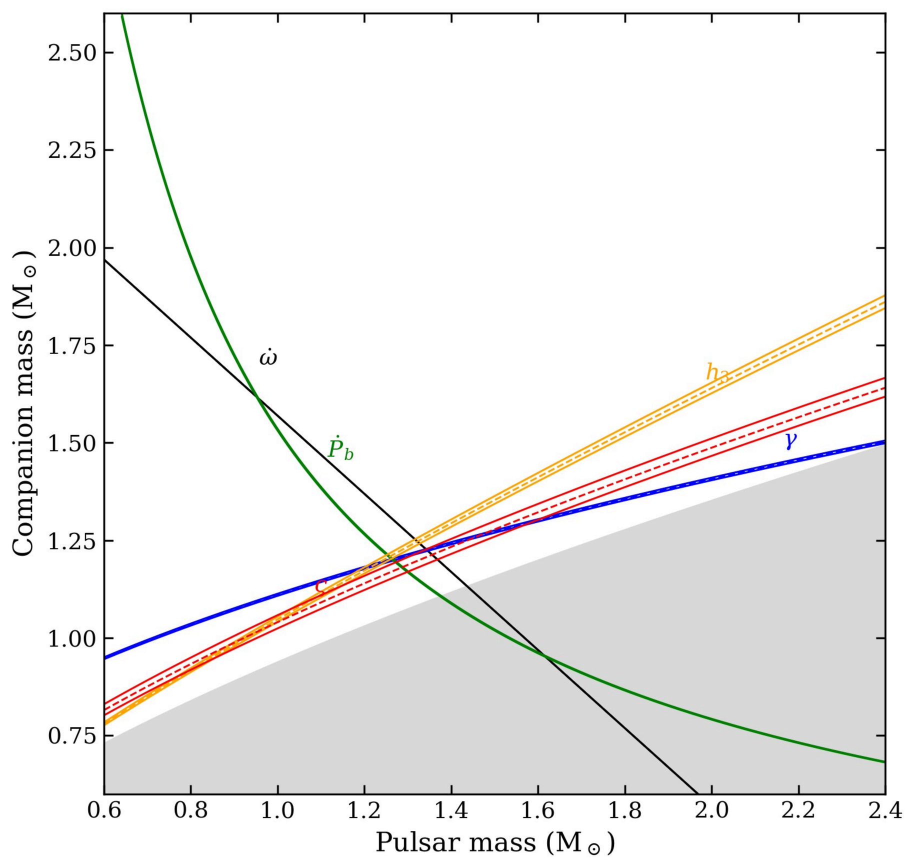


Fig. 4 *Updated mass–mass diagram from the DDGR + red-noise solution, showing five post-Keplerian constraints (ω̇, γ, Ṗb, and the orthometric Shapiro-delay parameters) with ±1σ bands; the grey region is excluded by sin i ≤ 1.*

In earlier studies, one could easily see the effect of a biased $\dot{P}_b$. This observed departure from the GR prediction can be attributed to the combined effects on the observed orbital decay of this system that are not resulting from kinematic biases. the robustness of any gravity test using the we derive from timing this pulsar is limited until we are able to better constrain the systematic effects that influence its measurement.

We have not been able to constrain these systematics in the latest timing analysis. We note a higher uncertainty in the proper motion of the pulsar and an unconstrained parallax.

4 Orbital decay and distance

The DDGR timing model can be reparametrised to yield the excess orbital-period derivative, $\Delta\dot{P}_b \equiv \dot{P}_{b,obs} - \dot{P}_{b,GR}$, which captures the kinematic contribution from proper motion (the Shklovskii effect) and differential Galactic acceleration along the line of sight. Under the assumption that the intrinsic orbital decay is entirely due to gravitational-wave emission as predicted by general relativity, we invert the kinematic bias equation to solve for the pulsar distance and propagate the uncertainties accordingly.

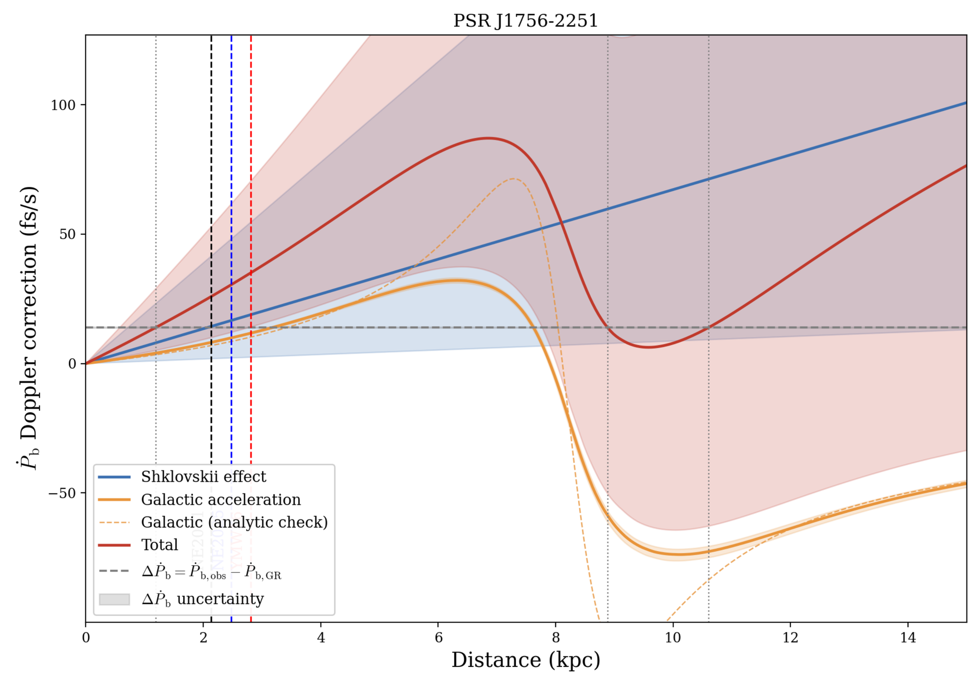
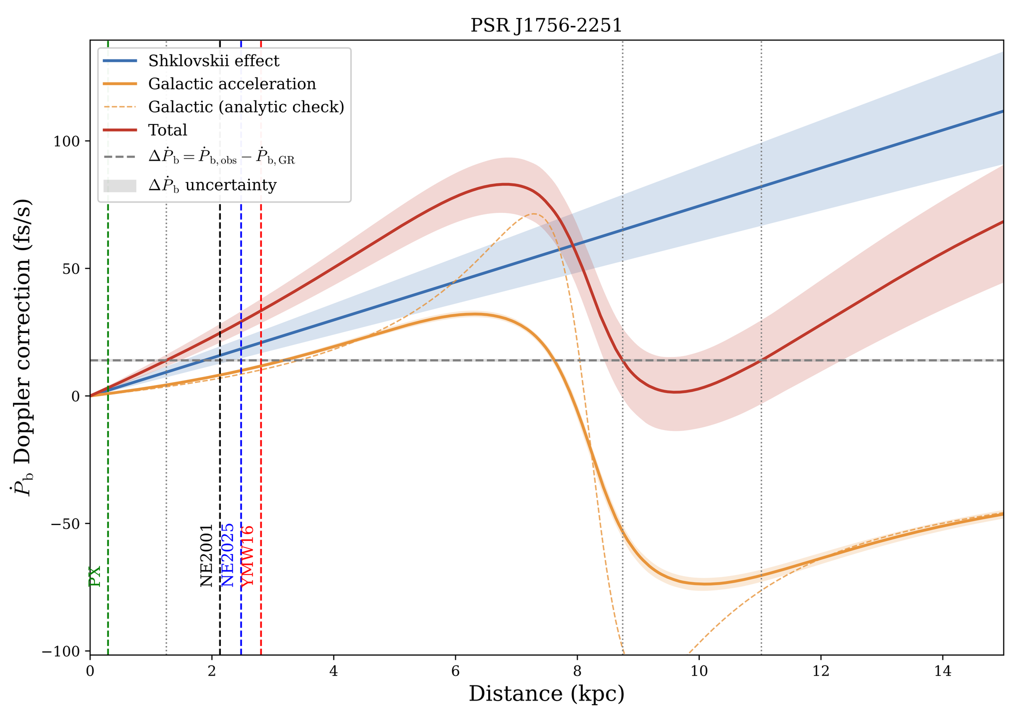


Fig. 5 *Kinematic Ṗb corrections as a function of distance. The Shklovskii and Galactic acceleration terms sum to the excess measured ΔṖb (dashed line), yielding a distance estimate; DM-model distances from NE2001, NE2025, and YMW16 are indicated.*

Result: The $\Delta\dot{P}_b$ distance is consistent with all three electron-density models (≈2–3 kpc) and supersedes the timing parallax, which remains unconstrained.

The dominant source of uncertainty in the total kinematic correction is the poorly constrained proper motion ($\mu_\delta = 9.9 \pm 7.0$ mas yr^{-1}) which enters the Shklovskii term as $\dot{P}_{b,Shk} \propto \mu_d^2$. Since $\dot{x}$ carries an independent imprint of the proper motion, we fit the TEMPO2 T2 binary model - parametrising the orbital inclination i and ascending node Ω alongside μ to obtain a significantly improved $\mu_\delta = 10.35 \pm 1.03$ mas yr^{-1}, a factor of ~7 reduction in the uncertainty, substantially tightening the kinematic Ṗb correction. The T2 parallax is in tension with DM-based distances and is not adopted.



This is a preliminary result, we are still analysing this method.

5 Pulsar position and solar wind

The proximity of PSR J1756−2251 to the ecliptic makes the timing sensitive to solar wind dispersion. In this analysis, times of arrival obtained during closest solar approach have been excised; a dedicated solar wind noise model will be incorporated in future iterations.

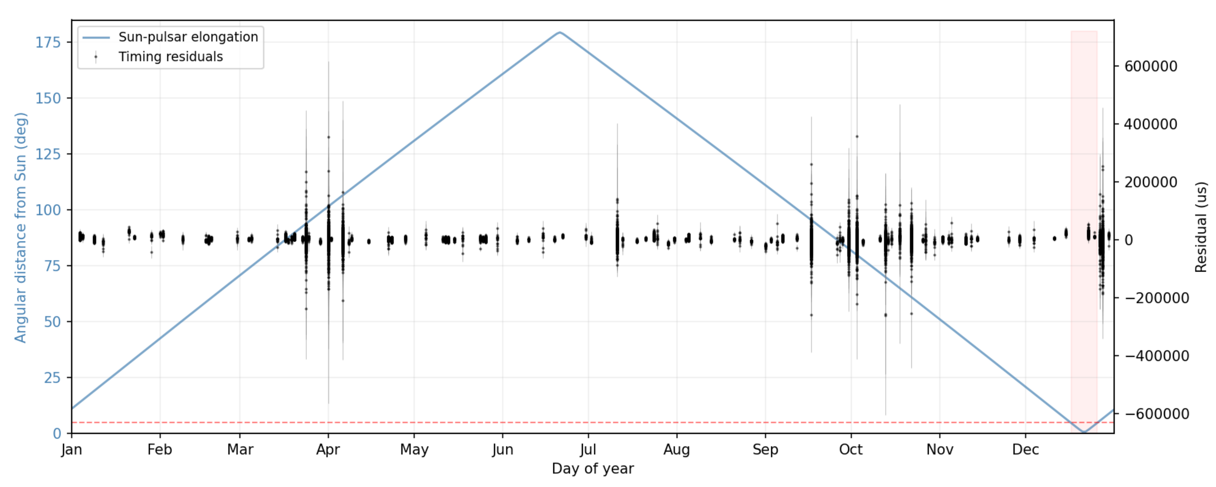


Fig. 7 *Annual angular separation of PSR J1756−2251 from the Sun, overlaid on the timing residuals. ToAs obtained within 5° of solar conjunction, where solar wind dispersion is significant, are highlighted.*

6 Seven years of polarimetric analysis with MeerKAT

Every MeerKAT epoch is examined for polarisation variability. Stacked L and V profile variations, differenced Stokes Q/U images and position-angle maps shows interesting changes that are not seen in the pulse intensity profile. The cause of this is still under investigation and any suggestions are welcome!

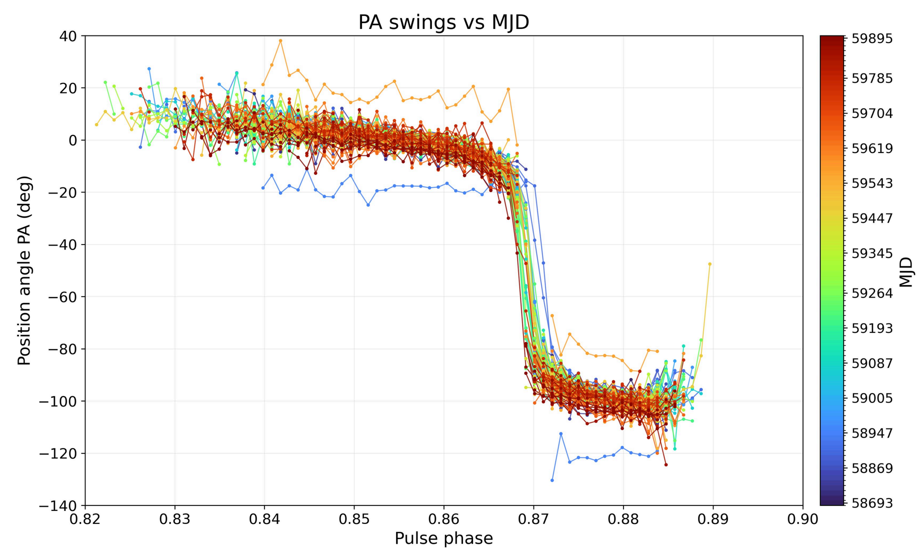


Fig. 8 *Faraday-corrected position-angle swings colour-coded by MJD, revealing a persistent orthogonal mode jump whose phase shows a marginal drift over seven years.*

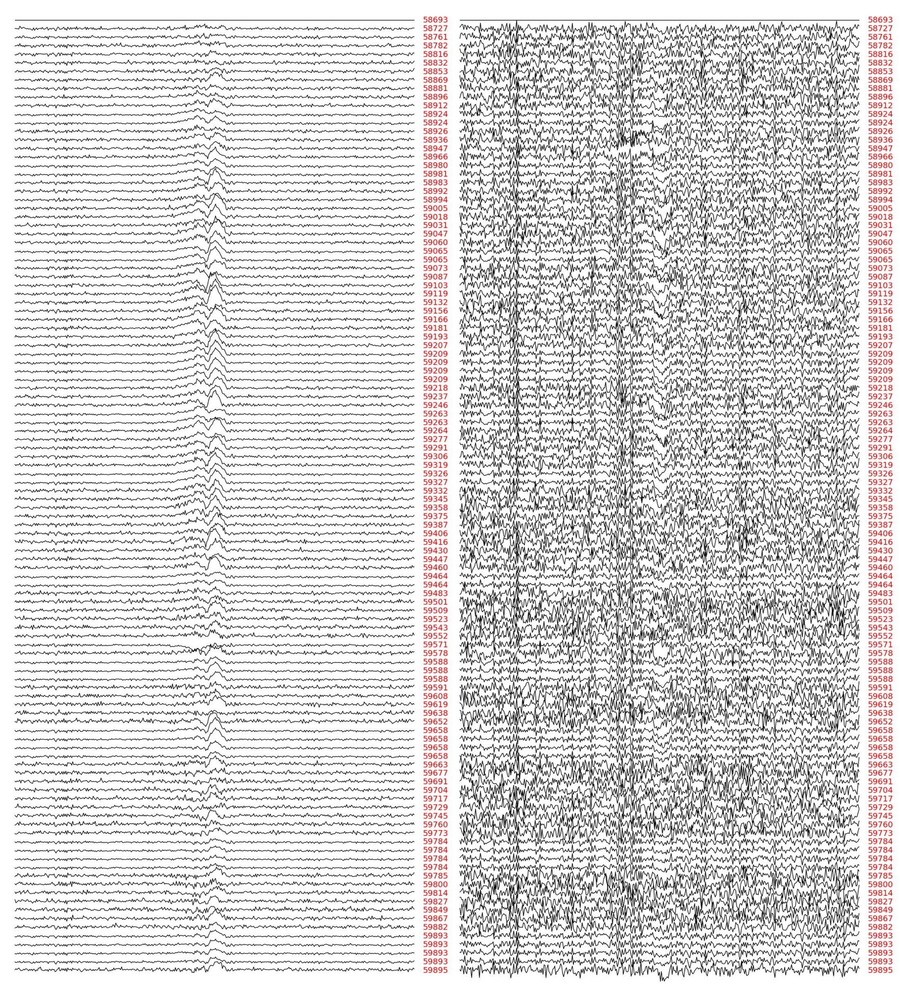


Fig. 9 *Stacked variance in linear (L) and circular (V) polarisation for MeerKAT campaign*

7 Gravity passes and the test goes on.

- Every post-Keplerian parameter is consistent with general relativity at current precision.
- The kinematic distance from ΔṖb is consistent with all DM-based electron-density models; the timing parallax remains unconstrained.
- Polarimetric variability - absent in total intensity raises open questions about the emission geometry.
- Continued Effelsberg monitoring will tighten the component masses and kinematic Ṗb correction year on year.
- VLBI astrometry is needed to independently confirm the proper motion and resolve the distance ambiguity.

KEY REFERENCES

[1] Faulkner et al. 2005, ApJ 618, L119
[2] Ferdman et al. 2014, MNRAS 443, 2183
[3] Damour & Deruelle 1986
[4] Hobbs et al. 2006, MNRAS 369, 655
[5] S. M. Kopeikin 1996 *ApJ* 467 L93

MeerKAT is operated by SARAO; Murriyang (Parkes) by CSIRO; the Effelsberg 100-m telescope by the MPIfR.

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